

PROBLEM STATEMENT

CIVICAI – Intelligent Public Grievance & Governance Assistant

Introduction

Public grievance redressal systems are an essential part of governance, enabling citizens to report civic issues such as sanitation problems, road damage, water leakage, electricity failures, drainage blockages, and public safety concerns. However, most traditional grievance systems still rely on manual workflows, making the process slow, inefficient, and difficult to scale. As urban populations continue to grow and cities become more complex, governments face increasing challenges in managing complaints effectively while ensuring transparency, accountability, and citizen satisfaction.

Existing Challenges

Current public grievance systems face several limitations:

- **Slow complaint processing** due to manual handling
- **No urgency-based prioritization**, causing critical issues to be delayed
- **Incorrect department routing**, leading to longer resolution times
- **Lack of transparency** in complaint tracking and updates
- **Duplicate complaints** increasing workload for authorities
- **Limited accessibility** for elderly and non-technical citizens
- **No intelligent analysis** of complaint patterns or recurring hotspots

These issues reduce public trust and create inefficiencies in governance.

Problem Gap

Most existing public grievance systems are designed only to register complaints, with little or no intelligence in how those complaints are processed. The absence of an integrated AI-driven mechanism creates inefficiencies at multiple stages of grievance handling.

Current systems fail to:

- **Understand complaints contextually**, leading to incomplete or inaccurate issue interpretation
- **Prioritize complaints dynamically** based on severity, urgency, and public impact
- **Route issues automatically** to the correct government department
- **Identify duplicate complaints**, increasing unnecessary workload and resource consumption
- **Support multi-modal reporting** such as voice and image-based complaints, limiting accessibility
- **Provide transparent and explainable responses**, reducing citizen trust in the system

As a result, critical civic issues often face delays, mismanagement and lack of accountability. This creates a major gap between citizen expectations for fast, transparent services and the current capabilities of traditional grievance systems.

Proposed Solution

To address these challenges, we propose **CivicAI**, an AI-powered governance assistant designed to automate and optimize the grievance management process.

CivicAI allows citizens to submit complaints through text, voice, or image inputs and uses IBM watsonx Orchestrate and IBM Granite models to:

- Analyze and understand complaints
- Classify them into relevant civic categories
- Assign urgency-based priorities
- Route them to the appropriate departments
- Generate structured complaint summaries
- Detect duplicates and identify civic hotspots
- Provide transparent next-step guidance

Objectives

The primary objective of **CivicAI** is to transform traditional grievance handling into an intelligent, automated, and citizen-centric governance system. It aims to bridge the gap between

citizens and public service departments by making complaint management faster, smarter, and more transparent.

The key objectives of CivicAI are:

- **To accelerate grievance resolution** by automating complaint analysis and prioritization
- **To minimize manual intervention** in complaint classification and triaging
- **To improve routing accuracy** by directing complaints to the appropriate government departments
- **To enhance transparency and accountability** through structured responses and clear reasoning
- **To improve accessibility** by supporting text, voice, and image-based complaint submission
- **To reduce duplicate complaints** and identify recurring civic hotspots for better planning
- **To strengthen citizen trust and engagement** by providing faster and more reliable public service interactions
- **To build a scalable governance solution** adaptable for municipalities, smart cities, and district administrations

Expected Impact

CivicAI aims to transform grievance handling from a reactive manual process into an intelligent and proactive governance system by:

- Improving complaint resolution efficiency
- Reducing administrative workload
- Increasing citizen trust and engagement
- Enabling smarter data-driven governance decisions

Conclusion

CivicAI is a step toward building smarter, faster, and more transparent governance systems. By combining AI, automation, and citizen-centric design, it has the potential to significantly improve public service delivery and strengthen civic administration.